

Classifying Life's Diversity

Earth has millions of living things. How do scientists keep them organized? They sort life into groups — like sorting a giant playlist. This guided edition walks you through it step by step, with word banks and sentence starters to help you.

 Georgia Standard S7L1

WHAT YOU'LL LEARN

- | | |
|---------------------------------------|--------------------------------|
| 01 What Makes Something Alive? | 05 The Six Kingdoms |
| 02 Why Sort Living Things? | 06 Tools for Sorting |
| 03 How Sorting Changed | 07 Naming Living Things |
| 04 The Levels of Sorting | 08 Review & Check |

CHAPTER WORD BANK

organism

classify

taxonomy

kingdom

species

nucleus

prokaryote

eukaryote

dichotomous key

scientific name

Tip: Each word is in **bold** the first time it appears. Look for the  boxes that explain what it means.

1.1 What Makes Something Alive?

First, a big question: how do we know if something is **alive**? A rock is not alive. A dog **is** alive. Scientists use a checklist to decide.

organism = any living thing. A dog, a tree, and a germ are all organisms.

To be alive, something must do **all six** things on this list. If it misses even one, it is not alive.

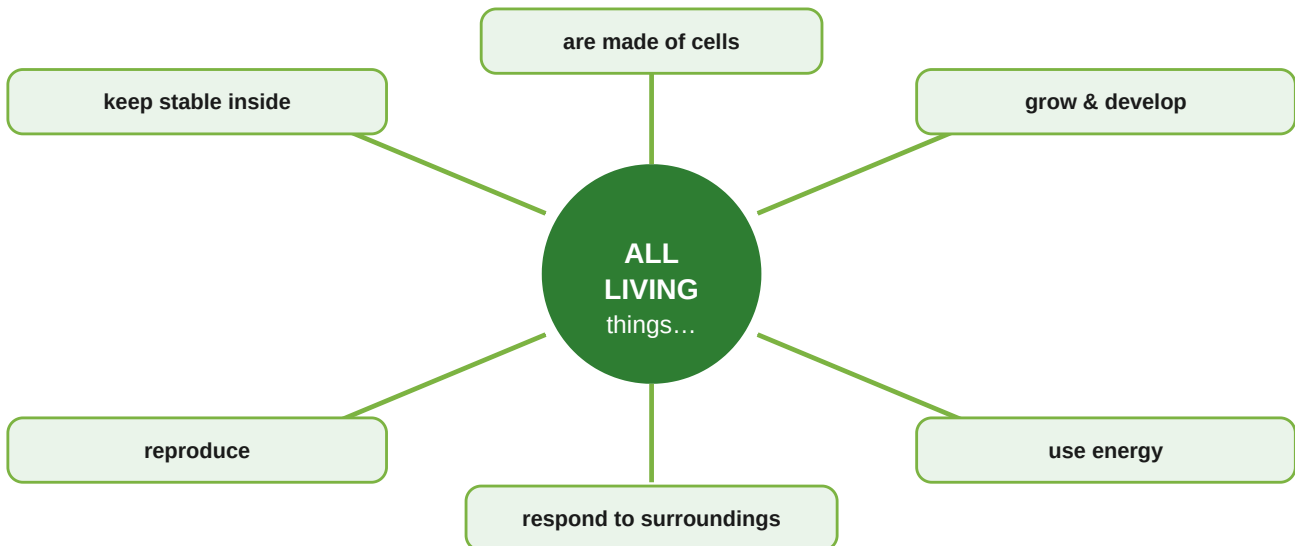


Figure 1.1 — The six things all living things do.

✓ Quick Check 1.1

1. A car uses energy and moves. Is a car alive? Why or why not?
2. Name **two** of the six things all living things do.

! Did You Know?

The biggest living thing on Earth is a fungus in Oregon. It is bigger than 1,600 football fields and may be over 2,400 years old!

1.2 Why Do Scientists Sort Living Things?

Imagine a store with everything in one big pile. No signs. No aisles. Finding milk would take forever! Scientists sort living things so they are easy to study.

classify = to sort things into groups.

taxonomy = the science of sorting living things.

Sorting living things helps in **three** ways:

- It **organizes** millions of living things.
- It **shows relationships** — things in the same group are related.
- It **helps us guess** — if an animal has fur and drinks milk as a baby, we already know a lot about it.

Sentence Starter

Scientists classify living things because _____.

Key Idea

Living things are grouped by the **traits they share**. The more traits two things share, the more closely related they are.

1.3 How Sorting Living Things Changed

Long ago, people sorted life in simple ways. As we built better tools — like the microscope — we learned more and changed the system.

When	Who	What They Did
Long ago	Aristotle	Used just 2 groups: plants and animals.
1735	Linnaeus	Made the system we still use. He sorted by looks and gave each a two-word name.
Later	Microscope users	Found tiny living things that were not plants or animals. New groups were added.
Today	Modern scientists	Use DNA to sort life into six kingdoms .

! Why It Keeps Changing

Old scientists could only sort by what they could **see**. Today, DNA lets us look **inside** the cell. Better tools = better sorting.

✓ Quick Check 1.3

1. How many groups did Aristotle use?
2. What tool today helps us sort living things more accurately?

1.4 The Levels of Sorting

Linnaeus's system has **eight levels**. The top level (Domain) is the **biggest** group. The bottom level (Species) is the **smallest**. As you go down, the groups get smaller and the living things become more alike.

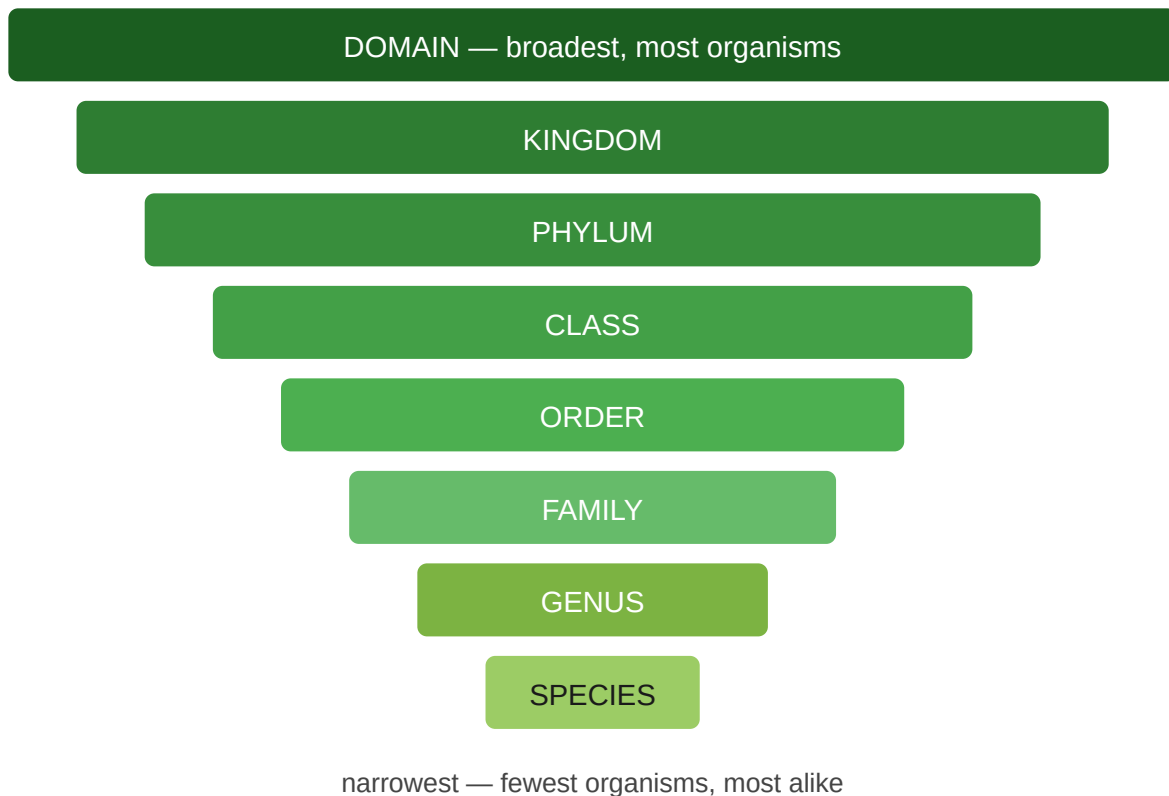


Figure 1.2 — The eight levels, from biggest to smallest.

★ Memory Trick

Say this silly sentence. The first letter of each word matches each level:

Dear **K**ing **P**hilip **C**ame **O**ver **F**or **G**ood **S**oup

= Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species.

📖 FILL IN THE LEVELS

Use the Memory Trick to fill the blanks in order:

D_____ → K_____ → P_____ → C_____ → O_____ → F_____ → G_____ → S_____

1.5 The Six Kingdoms of Life

At the top, all life is sorted into **six kingdoms**. To pick a kingdom, ask three questions: One cell or many? Does the cell have a nucleus? How does it get food?

nucleus = the control center of a cell that holds the DNA.

prokaryote = a cell with **no** nucleus.

eukaryote = a cell **with** a nucleus.

 Archaea 1 cell, no nucleus makes/absorbs <i>hot-spring microbes</i>	 Bacteria 1 cell, no nucleus absorbs food <i>strep, E. coli</i>	 Protists 1 or many cells varies <i>amoeba, algae</i>
 Fungi many cells absorbs food <i>mushroom, mold</i>	 Plants many cells makes own food <i>trees, moss</i>	 Animals many cells eats food <i>dog, human, ant</i>

★ The Big Split

The first question is: does the cell have a **nucleus**?

- **No nucleus** = Archaea and Bacteria (prokaryotes).
- **Has a nucleus** = Protists, Fungi, Plants, Animals (eukaryotes).

✓ Quick Check 1.5

1. Which two kingdoms have **no** nucleus?
2. Which kingdom makes its own food?
3. What kingdom is a mushroom in?

1.6 Tools for Sorting: The Key

To name an unknown living thing, scientists use a **dichotomous key**. It asks you yes/no questions until you find the answer.

dichotomous key = a list of either/or questions that helps you name a living thing. *Dichotomous* means "split into two."

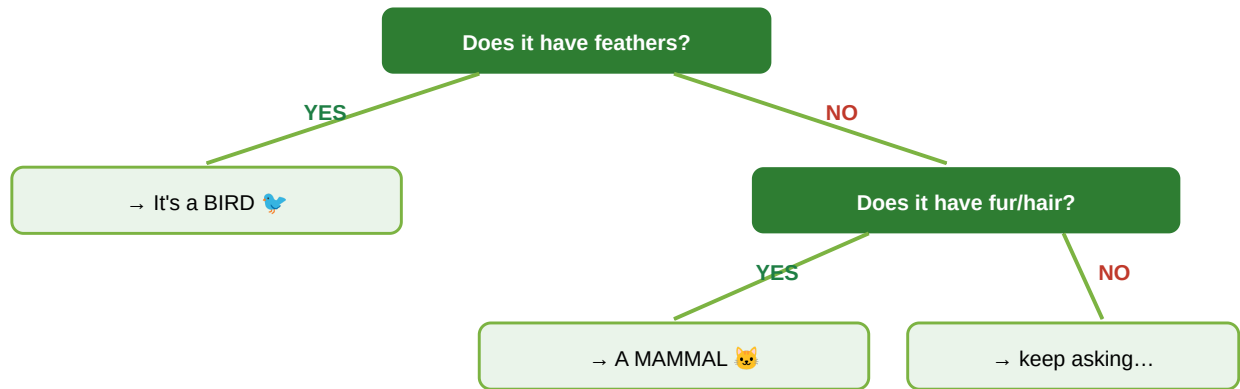


Figure 1.3 — A key. Each question has only two answers.

★ How to Use a Key — Steps

- Start at **step 1**.
- Read **both** choices.
- Pick the one that matches.
- Follow it until you reach a **name**.

1.7 Naming Living Things

A robin has a different name in every language. To stop the confusion, scientists give each species **one** special two-word name everyone uses.

scientific name = a two-word name made from the **Genus** and the **species**. Example: humans are *Homo sapiens*.



The 3 Rules

- Capitalize the **Genus** (first word).
- Lowercase the **species** (second word).
- *Italicize* the whole name (or underline it by hand).



Try It

Write the scientific name for a house cat. (Genus = *Felis*, species = *catus*.)

X Common Mix-Up

Don't capitalize the second word. Write *Homo sapiens* — never *Homo Sapiens*.

1.8 Review & Check

Great work! Here is everything in one place. Use it to study.

Big Idea	Remember This
Alive?	All living things do 6 things: made of cells, use energy, grow, reproduce, respond, stay stable inside.
Classify	Sort living things by shared traits .
History	Aristotle → Linnaeus → microscope → DNA.
Levels	Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species.
Six Kingdoms	Archaea, Bacteria, Protists, Fungi, Plants, Animals.
Naming	Two words: <i>Genus species</i> .

SHOW WHAT YOU KNOW — Sentence Starters

1. Two living things are placed in the same group when _____.

2. The classification system changed over time because _____.

★ Standard Check — S7L1

You can now **(a)** sort living things into groups by their shared traits, and **(b)** explain how sorting grew from simple "looks-based" groups into today's six kingdoms. 🎉